

# SMART INDIA HACKATHON 2026



## TITLE PAGE

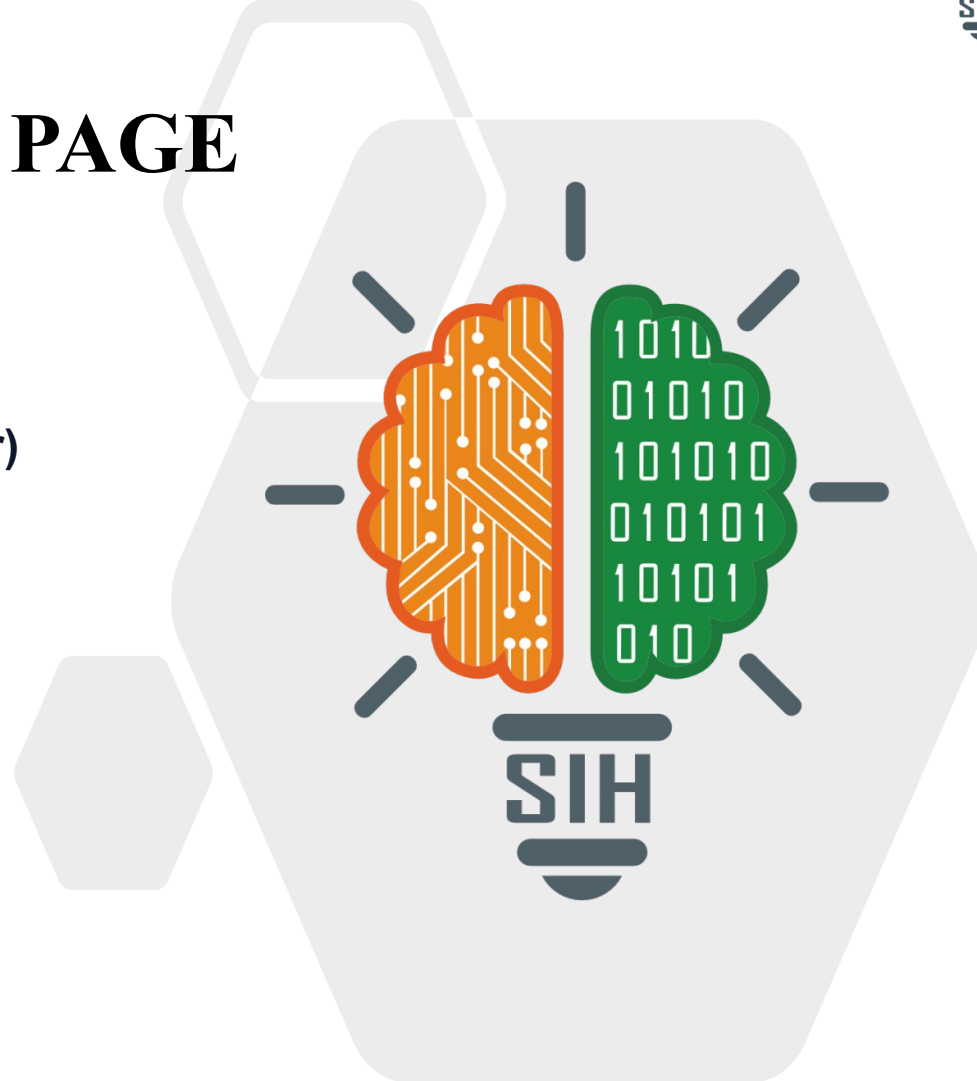
**Problem Statement ID –** [SIH26011](#)

**Problem Statement Title – 3D ULPIN (Bhu-Aadhaar)  
Generation, Vertical Cadastre & 3D Building  
Visualization Digital Twin**

**Theme –** Smart Automation / Geospatial Technology /  
Smart Cities

**PS Category –** Software

**Team Name (Registered on portal) –** [GeoVoxel The  
coders](#)



# GEO-CADASTRE 3D: Autonomous 3D ULPIN & Vertical Cadastre Digital Twin

## ⚡ Proposed Solution

- End-to-end autonomous geospatial pipeline that generates India's first production-ready 3D vertical cadastre, transforming 2D Bhu-Aadhaar (ULPIN) into legally actionable volumetric land parcels.
- Interactive 60 FPS WebGL digital twin built on Next.js 16 + Three.js, delivering real-time building inspection, floor isolation, and solar analysis in any browser.

## 🔍 Detailed Technical Solution

- Tier 1 (Unit-Level 3D ULPIN): Parses sanctioned TS-RERA architectural blueprints into individual flat/unit parcels ({ULPIN-2D}-FLxx-Uyyyy) with carpet area, facing direction, and Z-coordinates.
- Tier 2 (Floor-Level 3D ULPIN): Automated vertical cadastre ({ULPIN-2D}-FLxx) for all buildings via hybrid DSM-DEM height estimation with zero-discrepancy floor normalization.

## 🎯 How It Addresses The Problem

- Eliminates multi-ownership chaos where hundreds of apartment owners share a single 2D survey number — each unit now has its own unique 3D spatial identity.
- Creates undisputed legal 3D spatial parcels (Z-min, Z-max MSL) aligned with ISO 19152 LADM Part 3 and OGC CityGML 3.0 standards.

## 💡 Innovation & Uniqueness

- Zero-Discrepancy Floor Slab Engine: Proportional normalization guarantees  $\sum h_i \equiv H_{\text{building}}$  with 0.00m cumulative error across all 4,641 buildings.
- Client-Side 60 FPS Digital Twin: Full BatchedMesh / LOD rendering of 22,761 volumetric parcels + 238 km road network — no server GPU required.

### 🏛️ GOVERNMENT 2D BHU-AADHAAR (ULPIN)

**3621050103VUWK** (14-Digit DoLR Standard)

State: 36 · District: 21 · Mandal: 05 · Village: 0103 · Survey: 83/1

### TIER 1: UNIT 3D ULPIN (Sanctioned RERA Blueprints)

▶ -FL32-U3201

4BHK Luxury · 318 m<sup>2</sup> · East

▶ -FL32-U3202

3BHK Premium · 241 m<sup>2</sup> · West

▶ -FL32-U3203

3BHK Premium · 241 m<sup>2</sup> · North

▶ -FL32-ULOBBY

Common Area · Lift & Fire · 85 m<sup>2</sup>

### TIER 2: FLOOR 3D ULPIN (Automated Height Slicing)

▶ -FL32

Floor 32 · Z: 691.01m – 694.00m MSL

▶ -FL15

Floor 15 · Z: 639.75m – 642.75m MSL

▶ -FL08

Floor 8 · Z: 618.34m – 621.34m MSL

▶ -FL00

Ground · Z: 594.00m – 598.34m MSL

## STAGE 1 DATA INGESTION

- OpenStreetMap (Overpass QL)
- Copernicus GLO-30 DSM (AWS)
- SRTM GL1 DEM (OpenTopo)
- ISRO Bhuvan OGC Geoportal
- TS-RERA & Dharani Land DB

## STAGE 2 5-TIER HEIGHT ENGINE

- 1. Landmark / RERA Registry
- 2. DSM-DEM Zonal Statistics
- 3. Annular Buffer Sampling
- 4. Morphological Typology
- 5. Cluster Spatial Propagation

## STAGE 3 3D CADASTRE & ULPIN

- 14-Digit 2D Bhu-Aadhaar (NIC)
- Unit 3D ULPIN (-FLxx-Uyyyy)
- Floor 3D ULPIN (-FLxx)
- Zero-Discrepancy Normalizer
- ISO 19152 LADM / CityGML 3.0

## STAGE 4 3D DIGITAL TWIN

- Next.js 16 + React 19 + Three.js
- 60 FPS BatchedMesh / LOD
- Interactive Floor Isolation
- Solar & Flood Simulation
- 3D Flyovers & Road Network

4,641

Buildings Processed

22,761

3D Vertical Parcels

238 km

Road Network

12.5s

Pipeline Execution

60 FPS

Rendering Speed

### Technologies & Frameworks

- **Core Geospatial:** Python 3.10+, GDAL, Rasterio, Shapely, GeoPandas, PyProj, SciPy k-d trees, OpenTopography API.
- **Elevation & Remote Sensing:** Copernicus GLO-30 DSM (30m), SRTM GL1 DEM, ISRO Bhuvan OGC WFS/WMS, OSM Overpass QL, TS-RERA.
- **3D Digital Twin Platform:** Next.js 16 (Turbopack), React 19, Three.js (r185), React Three Fiber, BatchedMesh / WebGL 2.0.

### Methodology & Implementation

- **1. Data Ingestion:** Bounding box clipping, UTM 44N reprojection, OSM vector extraction, Bhuvan cadastral validation.
- **2. Height Engine:** DSM-DEM zonal diff + annular buffer terrain sampling + morphological typology + landmark registry.
- **3. 3D Cadastre:** 14-digit ULPIN, zero-discrepancy floor normalization ( $\sum h_i \equiv H$ ), dual-tier volumetric parcels.
- **4. Digital Twin:** Instanced rendering of 4,641 buildings + 238 km roads in 12.5s; interactive floor inspection.

## ✓ Analysis of Feasibility

- **Data Feasibility:** 100% open, authoritative sovereign data: Copernicus DEM, SRTM, ISRO Bhuvan, OSM, LGD codes. Zero recurring licensing fees.
- **Operational Feasibility:** Master pipeline processes 9 km<sup>2</sup> (4,641 buildings, 22,761 vertical parcels, 238 km roads) in 12.5 seconds on commodity hardware.
- **Technical Feasibility:** Built on validated open standards: 14-digit DoLR ULPIN, ISO 19152 LADM, OGC CityGML, and GeoJSON-3D.
- **Deployment Viability:** Client-side WebGL viewer runs at 60 FPS in standard desktop/mobile browsers — no GPUs or native installs required.

## ⚠ Potential Challenges & Risks

- ⚠ **Satellite Raster Noise:** Global 30m DSM/DEM rasters exhibit height distortion and ground clutter in dense high-rise clusters.
- ⚠ **Missing Floor Plans:** Over 95% of municipal multi-story buildings lack digitized CAD/BIM floor plans for unit-level mapping.
- ⚠ **Browser Memory Limits:** Rendering tens of thousands of volumetric property units simultaneously can exhaust WebGL resources.
- ⚠ **Elevation Drift:** Floor slab elevation rounding errors cause cumulative drift between computed slices and physical rooftops.

## ✓ Mitigation Strategies

- ✓ **5-Tier Fallback Matrix:** Combines annular terrain buffering, morphological typology rules, and crowd-verified landmark registries for robust accuracy.
- ✓ **Dual-Mode Architecture:** Automated floor-level vertical partitioning (-FLxx) as standard, seamlessly upgrading to unit-level (-FLxx-Uyyyy) when blueprints exist.
- ✓ **On-Demand Volumetric LOD:** Only the selected building is dynamically extruded into volumetric floor slices; others remain optimized batched meshes at 60 FPS.
- ✓ **Proportional Normalizer:** Strictly enforces  $\sum h_i \equiv H_{\text{building}}$  and snaps top floor ceiling to base + total\_height with 0.00m error.

## 🎯 Potential Impact on Target Audience

### • Dept. of Land Resources (DoLR) & Survey of India:

Production-ready blueprint to transition 2D Bhu-Aadhaar into a national 3D cadastral framework, ending airspace boundary overlaps.

### • Urban Local Bodies (GHMC / Smart City SPVs):

Unlocks 15–25% higher property tax collection by detecting unassessed floors, unauthorized penthouses, and FAR/FSI violations.

### • Homebuyers, Citizens & RWA Associations:

Grants undisputed legal title to specific vertical airspace parcels, replacing fragile Undivided Share of Land (UDS) agreements.

### • Banks, NBFCs & Mortgage Lenders:

Instant 3D geospatial collateral verification; completely eliminates fraudulent multi-mortgaging of the same apartment flat.

## ★ Benefits (Social, Economic, Environmental)

### 💰 Economic Benefits:

Eliminates billions of rupees in civil land litigation; accelerates bank loan approvals from weeks to minutes; enables transparent, view-based property valuations.

### 🏛️ Social & Governance:

100% transparency in vertical real estate ownership; prevents double-selling scams; empowers citizens with direct verification of sanctioned boundaries.

### 🚒 Disaster Management & Safety:

Floor-level flood inundation simulation (e.g. Durgam Cheruvu lake breach modeling) identifies submerged levels in real time for precision rescue operations.

### 🌿 Environmental & Urban Planning:

3D solar shadow path analysis enables accurate rooftop solar PV capacity estimation and micro-climate urban heat island mitigation strategies.

## Research & Reference Work

- **DoLR / NIC Technical Guidelines:** Dept. of Land Resources, MoRD, GoI: ULPIN / Bhu-Aadhaar Technical Specifications & Design Guidelines (2021–2024).
- **ISO 19152:2012 / 2024 (LADM):** Geographic Information — Land Administration Domain Model (LADM) — Part 3: 3D/4D Marine and Space Administration.
- **OGC Geospatial Standards:** City Geography Markup Language (CityGML 3.0) & 3D Tiles Community Standard for streaming massive geospatial models.
- **Survey of India & MoHUA:** National Geospatial Policy 2022 & National Urban Digital Mission (NUDM) Smart City guidelines.
- **Academic Research:** Biljecki et al. (2015): 'Applications of 3D City Models' (ISPRS IJGI); Stoter et al. (2020): '3D Cadastre in Practice'.
- **Authoritative Data Provenance:** Copernicus GLO-30 DSM (AWS), SRTM GL1 DEM (OpenTopography), ISRO Bhuvan OGC Geoportal, OSM (ODbL), TS-RERA Portal.

## Working Prototype & Open Source Repository

- **GitHub:** [github.com/lmad-81/SIH-26011](https://github.com/lmad-81/SIH-26011)
- **3D Viewer:** localhost:3000 (Next.js 16 + Three.js r185)
- **Pipeline:** `python scripts/pipeline.py --city hyderabad`

